

Review

A Survey of Machine Learning Approaches to IoT Security

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Abstract

The explosive growth of the Internet of Things (IoT) has expanded the attack surface across industrial systems, smart cities, healthcare, and homes, motivating a synthesis of recent advances in machine learning for IoT security and a clear statement of remaining gaps. This review conducted a systematic search of MDPI, IEEE Xplore, Nature, ScienceDirect, and SpringerLink for publications from 2023 to 2025, screening them for domain relevance and organizing findings into a taxonomy of ML methods, threat types, and deployment contexts, with particular attention to datasets, edge constraints, and privacy considerations. We find that the field is shifting from signature-based detection to supervised and deep learning approaches that report high accuracy on benchmark traffic, while federated learning enables privacy-preserving, distributed intrusion detection with near-real-time edge performance. Across domains, prevalent threats include DDoS, unauthorized access, and malware; persistent challenges include device heterogeneity, rapid exploit weaponization, nonstandardized evaluation, concept drift, adversarial/poisoning risks, and governance and privacy constraints that hinder real world rollouts. We conclude that ML materially strengthens IoT resilience but requires rigorous, industry-scale validation, lightweight and explainable models, protocol-aware designs, robust federated aggregation, and SDN/NFV orchestration; we outline benchmark and deployment priorities to translate laboratory gains into operational security.

Keywords: Internet of Things; machine learning; intrusion detection systems; privacy protection

1. Introduction

In today's economy, interconnection and the transfer of large amounts of data have become fundamental. The main elements of this evolving ecosystem are sensors. We can find them integrated inside our everyday electronic devices or into highly specialized devices that are capable of monitoring everything from environmental parameters to the physiological indicators of humans or animals, contributing to a better quality of life. Smart devices communicate and exchange data over the internet or private networks, outlining the notion of the Internet of Things (IoT), which can be defined as an extended open network of smart devices that can share data and resources, are capable of self-organizing, and react to changes in their external environment. While such interconnectedness creates novel pathways for progress, it concurrently generates intricate security challenges that require systematic analysis and the implementation of new solutions, from the lowest levels of this architecture to the cloud level (Figure 1).



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According to [1], in 2024, globally, 18.8 billion IoT devices were connected, with this number expected to grow to an estimated 40 billion devices by 2031. The large-scale collection of diverse information, from human health parameters in healthcare systems to data from critical infrastructure elements, creates an increasing need for secure storage and processing that maintains confidentiality and privacy. Inadequate security in the context of IoT can pose significant risks, including exposing personal data, compromising critical infrastructures, and diminishing public trust in these technologies.

The urgency of addressing IoT security gaps is underscored by recent threat-landscape statistics. According to the Positive Technologies 2024 industrial IoT threat report, Advanced Persistent Threat (APT) attacks accounted for 19% of all manufacturing sector breaches in the first three quarters of 2024 [2], while ransomware targeting industrial control systems caused an average of 24 days of operational downtime per incident [3]. The substantial economic impact of 2024 cybersecurity breaches in IoT-enabled critical infrastructure is reflected in an average cost of \$5.56 million per incident, with industrial sectors facing expenses 13% higher than the cross-industry average [4]. The attack surface expansion is accelerating faster than defensive capabilities, vulnerability weaponization timelines have compressed from weeks to mere hours, as evidenced by CVE-2023-1389 being integrated into active botnets within days of public disclosure [5,6]. With IoT attacks increasing by 107% [7] year-over-year and a majority of organizations recognizing escalating threats to IoT devices [8], yet only 18% [9] of organizations having fully adopted AI cybersecurity tools, the gap between threat sophistication and defensive readiness warrants systematic investigation of ML-based defensive approaches across diverse IoT deployment contexts.

These escalating threats highlight why established security paradigms, such as signature-based intrusion detection, rule-centric firewalls, and static access control lists, encounter fundamental constraints in IoT ecosystems. The scale of typical installations (thousands to millions of heterogeneous endpoints), device diversity across protocols (Zigbee, Z-Wave, MQTT, CoAP, LoRaWAN) and accelerated weaponization dynamics (exploitation within hours of disclosure) collectively render manual rule curation and signature upkeep untenable. Simultaneously, IoT devices face stringent computational and energy limitations: battery-powered edge sensors with kilobytes of memory and megahertz-class processors lack the capacity for resource-intensive signature matching or complex cryptographic operations. The distributed topology of IoT systems, coupled with intermittent connectivity and extended operational lifetimes frequently spanning years without firmware maintenance, further reduces the viability of centralized security management and creates enduring exposure windows where signature-based defenses offer limited mitigation.

In light of these constraints, machine learning represents a qualitatively distinct security paradigm well aligned with IoT's operational limitations and rapidly evolving threats. Unlike rule-based defenses requiring explicit attack signatures, ML approaches infer patterns directly from IoT traffic, enabling detection of novel attack variants, including zero-day exploitation, with reduced reliance on human-authored rules and offering a counterweight to accelerated weaponization timelines. ML's scalability maps naturally onto IoT deployments with large device populations and high-throughput data streams, while learning-based methods can develop shared behavioral representations across diverse device classes and protocols, abstracting away low-level implementation differences. Recent progress in lightweight model design, federated learning, and edge-oriented inference optimization has improved the practicality of deploying ML on resource-constrained endpoints.

To better comprehend the scale of existing vulnerability issues within IoT ecosystems, Table 1 presents a series of attacks targeting the IoT domain between 2020 and 2025. Three

interrelated factors emerge from these incidents. First, weak authentication credentials, rapid vulnerability weaponization (disclosure-to-exploitation compressed to hours, as with Arcadyan CVE-2021-20090 and TP-Link CVE-2023-1389), and pervasive end-of-life devices without vendor support create persistent attack vectors exploited by both commodity botnets and nation-state actors such as Volt Typhoon [10]. Second, attack impacts span multiple dimensions: large-scale DDoS operations from compromised cameras and routers, privacy violations from direct device breaches (Verkada [11]), and cross-domain cascading failures where IoT compromise triggers physical-world consequences, as demonstrated when the Viasat/AcidRain modem sabotage [12] disrupted wind turbine telemetry. Third, the threat landscape has evolved from commodity DDoS botnets toward strategic operations leveraging edge devices for persistent infrastructure access, with 2024–2025 campaigns against AVTECH [13] and GeoVision [14] demonstrating systematic targeting of abandoned device ecosystems.

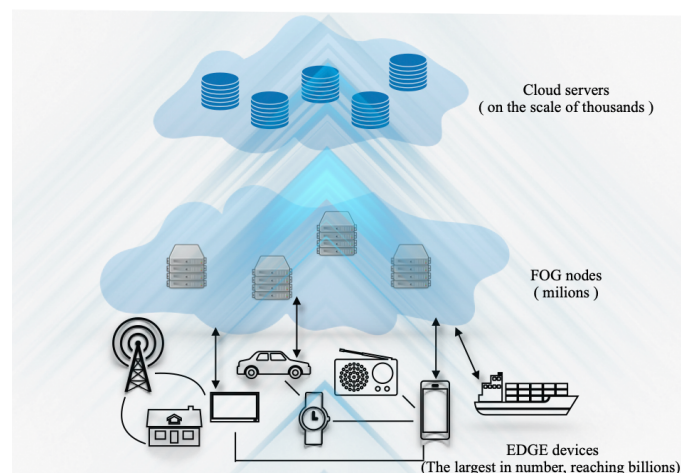


Figure 1. Conceptual architecture in IoT infrastructure.

While existing surveys [15–17] have examined ML techniques for IoT security, they present approaches generically without differentiating application contexts: none systematically maps ML method selection to the divergent threat profiles, operational constraints, and regulatory requirements of specific IoT deployment domains, nor do they quantify the gap between FL theoretical proposals and practical library-based implementations. This review advances existing taxonomies through three equally weighted contributions: (1) Domain-specific requirement mapping across smart homes, smart cities, healthcare, and industrial IoT, demonstrating that optimal ML approaches vary substantially by deployment context, with explicit rationale for each domain’s method selection; (2) Maturity-gap quantification of recent advances (2023–2025) in federated learning, explainable AI and hybrid architectures, evidenced by the finding that only 22/104 FL-IDS studies employ actual FL libraries [18]; and (3) a practitioner-oriented research roadmap organized into immediate, mid-term, and long-term implementation horizons, providing actionable recommendations grounded in domain-specific evidence rather than broad future work statements.

Machine learning mitigates these challenges through capabilities that map directly onto IoT security requirements: anomaly detection [19] identifies deviations from learned baselines, offering protection against threats lacking known signatures; continuous learning enables detection logic to adapt as environments evolve and adversarial techniques change; federated learning supports cross-organization threat intelligence without disclosing sensitive datasets, particularly relevant under healthcare and industrial confidentiality constraints; explainability approaches (SHAP, LIME) translate opaque outputs into action-

able alerts meeting auditing and compliance expectations; and edge-oriented techniques (model compression, quantization, hardware-aware acceleration) enable on-device inference under tight compute and energy budgets. These capabilities, combined with scalable pattern extraction from high-dimensional IoT data streams, position machine learning as a practical imperative for IoT security, as demonstrated by recent implementations reporting sub-millisecond edge inference with strong detection performance. The remainder of this work examines how current ML techniques satisfy these requirements across multiple IoT application domains, assesses their effectiveness against evolving threat vectors, and delineates the residual barriers to operational deployment.

Table 1. IoT security incidents between 2020 and 2025.

Year	Attack Name	Targeted Domain	Impact
2020	Mozi IoT botnet surge [20]	SOHO routers, DVRs, gateways	Large-scale DDoS capability; persistent infections across consumer IoT devices
2021	Verkada camera breach [11,21]	Cloud-managed security cameras (enterprise/industrial)	Access to 150 k live cameras; privacy exposure across hospitals, prisons, schools, firms
2021	Arcadyan router auth bypass (CVE-2021-20090) weaponized [22]	Home/SOHO routers (multiple brands using Arcadyan firmware)	Rapid global recruitment of routers into DDoS botnets days after disclosure
2021–2022	Hikvision camera RCE (CVE-2021-36260) exploited [23]	IP CCTV cameras	Camera takeover, lateral movement footholds, botnet growth in enterprise networks
2022	Viasat KA-SAT modem sabotage (“AcidRain”) [12]	Satellite broadband modems (consumer/industrial)	Tens of thousands modems disabled; 5800 wind turbines lost remote connectivity
2023	TP-Link Archer AX21 RCE (CVE-2023-1389) mass-exploited [5,6]	Home/SOHO Wi-Fi routers	Widespread router conscription, sustained DDoS firepower across multiple botnets into 2024
2023–2024	Volt Typhoon & KV-botnet [10]	SOHO routers (Cisco/NETGEAR, often end-of-life)	Pre-positioning in critical infrastructure; U.S. takedown and notifications January 2024
2024	AVTECH IP camera zero-day campaign [13]	IP cameras/NVRs (AVTECH)	Camera fleets co-opted into DDoS botnets; risk to critical environment organizations
2025	GeoVision EoL camera exploits (CVE-2024-6047, -11120) [14]	Legacy/EoL GeoVision IoT cameras	DDoS botnet expansion using abandoned devices with no vendor support

2. Methodology

This literature review aims to systematically examine contemporary research developments and emergent paradigms in machine learning (ML) applications for Internet of Things (IoT) security. The present analysis seeks to critically assess the efficacy of ML methodologies in mitigating security vulnerabilities, identify existing gaps in the current body of knowledge, and elucidate novel approaches that contribute to strengthening IoT system resilience.

Additionally, this review provides a taxonomic classification of ML techniques according to their domain-specific implementations within the IoT security landscape.

2.1. Search Strategy and Data Sources

- Database selection

A comprehensive literature search was conducted across five prominent academic databases selected for their comprehensive coverage of computer science, engineering and interdisciplinary IoT research:

- MDPI (Multidisciplinary Digital Publishing Institute)
- IEEE Xplore Digital Library
- Nature Publishing Group
- ScienceDirect
- SpringerLink

- Temporal scope

Publications from January 2023 to December 2025 (inclusive) were selected.

- Search keywords

The search strategy employed Boolean combinations of the following keyword sets:

- IoT security terms:
 - * “Internet of Things” OR “IoT”
 - * “security” OR “intrusion detection” OR “anomaly detection” OR “threat detection” OR “cybersecurity” OR “attack detection”
- Machine learning terms:
 - “machine learning” OR “deep learning” OR “neural network” OR “CNN” OR “LSTM” OR “random forest” OR “federated learning” OR “supervised learning” OR “unsupervised learning”
- Domain-specific terms:
 - “smart home” OR “smart city” OR “healthcare IoT” OR “industrial IoT” OR “IIoT”

- Initial results

The search queries yielded approximately 120,000 candidate publications across all databases. These were progressively filtered through a structured screening process; Table 2 documents the selection funnel.

Table 2. Paper selection funnel.

Screening Stage	Papers Remaining	Action/Exclusion Reason
Initial database search	~120,000	MDPI: ~12,000; IEEE Xplore: ~48,000; Nature: ~9000; ScienceDirect: ~28,000; SpringerLink: ~23,000. Searches run independently per database using identical keyword sets; duplicate records arising from cross-indexing removed at the deduplication stage
Date filter (January 2023–December 2025)	~9500	Excluded publications outside the 2023–2025 window
Deduplication	~7200	Removed duplicate records appearing across multiple databases
Title and abstract screening	~420	Retained only papers addressing ML/DL methods for IoT security; excluded generic network security or non-IoT studies
Full-text eligibility assessment	~90	Applied inclusion/exclusion criteria (see Sections 2.2 and 2.3); excluded review papers without novel contributions and studies lacking empirical evaluation on IoT-specific datasets
Final included papers	40	Primary research studies covering at least one of the four IoT domains (smart home, smart city, healthcare IoT, industrial IoT) with quantified ML performance metrics

Domain relevance screening further required that each retained paper address a specific IoT deployment context rather than general network environments: studies evaluating solely on generic datasets (e.g., KDD Cup 99) without IoT-specific validation were excluded at the full-text stage.

2.2. Inclusion Criteria

- Publication date between 1 January 2023 and 31 December 2025
- Focus on ML/DL applications for IoT security (not general network security)
- Empirical evaluation on IoT-specific datasets or real-world deployments
- Full text availability in English or Romanian language

2.3. Exclusion Criteria

- Publications before 2023 or after 2025
- Studies focusing solely on non-IoT network environments
- Review papers without novel contributions

2.4. Data Extraction and Classification Framework

- Extracted attributes
 - ML technique taxonomy: supervised, unsupervised, reinforcement learning, federated learning, hybrid approaches
 - IoT application domain: smart home, smart city, healthcare IoT, industrial IoT
 - Addressed threats: DDoS, unauthorized access, malware, ransomware, data tampering
 - Performance metrics: accuracy, precision, recall, F1 score, inference latency, memory footprint
 - Deployment considerations: edge feasibility, privacy mechanisms, explainability
- Classification methodology
Machine learning techniques were categorized according to their implementation across different IoT domains, enabling systematic evaluation of domain-specific effectiveness.

2.5. Source Classification

The 67 references in this review fall into two distinct categories. Primary research sources (40 peer-reviewed papers and surveys) are academic publications included via the selection process described above; they provide the ML performance evidence analyzed in Sections 3–6. Background sources (20 references) comprise vendor threat-intelligence reports (Positive Technologies, IBM X-Force, Varonis), industry statistics (IoT Analytics, Cobalt, OneKey), government advisories (CISA), security-incident analyses (Akamai, SentinelOne) and web-based industry trend sources; these are cited exclusively to contextualize the threat landscape in Section 1 and are not used as evidence for ML performance claims. The remaining seven bibliography entries are market-research reports retained for background context but not directly cited in the text.

2.6. Synthesis and Analysis Approach

The data analysis focused on identifying trends, existing research gaps, and emerging opportunities through:

- Quantitative synthesis
Aggregating performance metrics across similar approaches to identify the best-performing paradigms.
- Qualitative analysis

Extracting common challenges, limitations, and future directions stated across multiple studies.

3. IoT Security Landscape

While existing surveys present IoT security requirements generically [16,17], the literature reveals that optimal ML approaches vary across application contexts due to divergent threat profiles, operational constraints, and regulatory requirements [24–26]. IoT devices are characterized mainly by limited computing power, memory, and energy resources, rendering the implementation of traditional, resource-intensive security protocols difficult [27]. This section provides a systematic domain-centric analysis of four critical IoT deployment contexts [27,28]. Unlike Alwahedi et al. [16] and Ali et al. [17], who present IoT security requirements at a uniform abstraction level, the analysis below treats each domain as a distinct constraint envelope and asks why specific ML paradigms succeed or fail under that envelope.

3.1. Smart Home IoT

Smart home environments face unique security challenges due to extreme device heterogeneity, user privacy expectations, and resource constraints. The threat landscape is dominated by DDoS attacks exploiting weak authentication for botnet recruitment and unauthorized access via brute-force methods [13,20,29]. Recent threat intelligence documents IoT-to-IoT attack scenarios where 67 compromised devices function as both attackers and victims within the same home network [29], illustrating lateral movement risk in heterogeneous deployments.

Security requirements identified in the literature prioritize privacy preservation avoiding centralized data collection [30,31], user interpretability enabling non-technical occupants to understand security decisions [30,32], edge-feasible deployment with sub-second latency [30,33] and multi-protocol support addressing Zigbee, Z-Wave, Wi-Fi and Bluetooth coexistence [24].

Predominant approaches favor classical ML (Random Forest, Decision Trees) achieving high accuracy with compact model sizes suitable for consumer hardware [30,34,35]. Feature selection reduces dimensionality from 41 to 18 features without accuracy loss [35], while CTGAN synthetic augmentation addresses class imbalance in residential traffic [30]. Research gaps persist in protocol heterogeneity, where most studies evaluate a single stack [36] and cross-device validation, where models trained on narrow device subsets lack generalization evidence across heterogeneous smart-home fleets [29].

The dominance of tree-based classical ML in this domain is structural rather than coincidental. Three coupled constraints select for it: consumer-gateway hardware (ARM CPUs, sub-watt budgets, no GPU) excludes attention and full-recurrent architectures whose inference latency exceeds residential threat-response windows; non-technical occupants require interpretability that Gini-based feature importance provides natively, without the SHAP/LIME overhead that gateway hardware barely tolerates; and homogeneous residential traffic, dominated by a handful of protocol families, does not exercise the automatic feature extraction that justifies CNN complexity in heterogeneous fleets. The marginal accuracy lift of CNN-GRU (99.39%) [33] over Random Forest (99.88%) [35] on smart-home benchmarks therefore comes with a net structural loss in interpretability and deployability.

3.2. Smart City IoT

Smart city deployments present unprecedented scalability challenges, processing data from millions of heterogeneous sensors across transportation, utilities and public safety infrastructure [17,31,37]. The attack surface spans critical municipal systems with cascad-

ing failure risks, as demonstrated by DDoS attacks disrupting intelligent transportation networks and energy distribution grids [31,38].

Security requirements demand privacy-preserving architectures compliant with GDPR and local regulations, scalability to millions of distributed devices [31], real-time response with sub-2 ms latency for time-critical applications [31] and multi-protocol support spanning IPv4/6, LoRaWAN, MQTT and CoAP [17].

Federated learning emerges as a necessary paradigm, achieving 94–98% accuracy across 10–150 clients with CNN architectures while maintaining edge feasibility (1.4 ms/sample, <72 °C on Raspberry Pi 5) [31]. Dynamic aggregation (FedMADE) improves minority attack detection by up to 71% with minimal overhead (4.7% latency increase) [38]. However, few FL-IDS studies validate with production frameworks [18] and the field lacks standardized city-scale benchmarks combining realistic traffic dynamics with multi-stakeholder privacy constraints [16,18].

Federated learning is structurally necessary in this domain rather than merely advantageous. GDPR data-locality provisions and municipal data-sovereignty requirements prohibit the centralized aggregation that classical centralized ML assumes; the scale of millions of sensors across multiple stakeholders makes a single training corpus operationally infeasible even where legally permitted; and the heterogeneity of municipal traffic, transit, utilities, and public safety, produces non-IID partitions that demand client-side training to preserve local distribution fidelity. The choice between FedAvg and dynamic aggregation (FedMADE [38]) becomes the second-order question; the first-order decision against centralized ML is forced by the constraint envelope itself.

3.3. Healthcare IoT

Healthcare IoT systems demand exceptional security due to patient safety implications and physiological data sensitivity [26,39]. The threat landscape encompasses data tampering and spoofing, unauthorized device access and DDoS attacks compromising system reliability [39].

Security requirements prioritize multi-objective security integrating anomaly detection, integrity verification and access control within unified frameworks [39], explainability for clinical and regulatory audit [30], real-time response for continuous vital sign monitoring [39] and privacy-preserving federated learning for distributed detection [40].

Hybrid architectures (CNN-LSTM-VAE) achieve multi-objective security: anomaly detection ($F_1 \approx 0.943$), integrity verification ($F_1 \approx 0.928$, TPR 94.6%, FPR 3.7%) and adaptive access control (96.1% accuracy) through Trust-Aware Controllers [39]. Edge deployment on Raspberry Pi 5 and Jetson Nano achieves 60% model size reduction with <1% accuracy degradation via quantization and TensorRT optimization [39]. Critical gaps include healthcare-specific dataset scarcity, with most studies relying on general IoT corpora [26] and limited formal privacy guarantees in cross-institutional federated learning [18].

The case for hybrid multi-objective deep learning in healthcare IoT rests on a constraint that simpler architectures cannot satisfy: anomaly detection, integrity verification and adaptive access control must be reconciled within a single decision frame, because independent classical-ML models produce inconsistent thresholds and conflicting boundaries that propagate to clinical workflows. The CNN-LSTM-VAE + Trust-Aware Controller architecture [39] achieves this by sharing representation across objectives, paying the parameter-count premium against the regulatory and patient-safety cost of inconsistent multi-channel decisions. Edge-deployment feasibility on Raspberry Pi 5/Jetson Nano via quantization and TensorRT [39] closes the loop: the architectural complexity that the regulatory constraint demands fits within the latency budget that continuous monitoring imposes.

3.4. Industrial IoT

Industrial IoT environments support critical infrastructure and operational technology networks where cyberattacks cause physical-world consequences [41]. The IIoT market is expanding rapidly across manufacturing, logistics and smart factory deployments [42–45], continuously enlarging the attack surface exposed to sophisticated adversaries. IIoT systems face sophisticated ransomware causing significant operational disruption and breach costs exceeding USD 5.56 million [3,46], while APT attacks account for 19% of manufacturing breaches [2,4]. Vulnerability weaponization timelines have compressed from weeks to hours [13,22].

IIoT privacy concerns center on intellectual property protection: workplace monitoring tools expose employees and sensitive operational data [25], while proprietary manufacturing processes captured by IIoT sensors constitute valuable targets for industrial espionage [25,47]. Trust mechanisms must verify sensor reliability and preserve data integrity throughout automated decision-making pipelines [48].

Security requirements emphasize adversarial robustness against nation-state APT campaigns [2,10], real-time performance for continuous threat detection [30,49], protocol-aware detection addressing application-layer vulnerabilities [25,50] and operational continuity through resilient architectures [51,52].

Predominant approaches include attention-based architectures with multi-head attention [52,53], hybrid CNN-GRU models [33,54] and meta-heuristic optimization [52,55]. Research gaps include the disconnect between laboratory evaluation and operational deployment [25], the need for enhanced adversarial robustness testing [52] and expansion beyond network-layer to application-layer security [27,56].

The preference for attention-based and ensemble architectures in IIoT, despite their training cost, follows from the threat model rather than from accuracy chasing. APT campaigns and multi-stage ransomware exhibit temporal dependencies, reconnaissance, lateral movement, payload deployment, that feature-engineered classical ML cannot capture without per-attack hand-tuning. Multi-head attention captures these dependencies through dynamic feature focus [52], and the 14,800-s training cost [57] that would disqualify the architecture in consumer or healthcare contexts is absorbed by OT maintenance windows that already accommodate multi-hour change cycles. The remaining open question, adversarial robustness under nation-state APT, is what blocks current academic implementations from operational deployment, not the inference latency itself.

The pattern of N/R cells in FPR, latency and energy/resource columns is itself a finding: of the 17 reviewed implementations, only one [39] reports FPR explicitly, six report latency, and one reports thermal/energy data. As Table 3 illustrates, this near-universal absence of non-accuracy reporting, not the underlying methods, is the binding constraint on practitioner method selection.

Table 3. Multi-metric comparison of representative ML-based IoT security implementations (2023–2025). N/R = not reported.

Domain	Study	ML Approach	Acc.	FPR	Latency	Resource/Energy	Dataset
Smart home	Hasan [30]	RF + CTGAN augmentation	97–100%	N/R	0.01–0.57 s	2.73–1510 kB	CICIoT2023
Smart home	Nawaz [35]	XGBoost + feature selection	99.88%	N/R	N/R	41 → 18 features	CICIoT2023
Smart home	Wang [33]	CNN-GRU hybrid	99.39%	N/R	8 ms/sample	N/R	Custom IoT
Smart home	Karakaya [34]	Hybrid classical ML	>99%	N/R	N/R	N/R	NSL-KDD/CIC
Smart home	Alabbadi [32]	1-D CNN + XAI (LIME)	up to 100%	N/R	N/R	N/R	IoT-specific
Smart city	Albanbay [31]	Federated CNN (10–150 clients)	94–98%	N/R	1.4 ms/sample	RPi 5 (<72 °C)	CICIoT2023
Smart city	Sun [38]	FedMADE (dynamic agg.)	F_1 0.981	N/R	+4.7% overhead	N/R	CICIDS
Smart city	El-Sofany [19]	SVM/classical ML	94–96%	N/R	N/R	N/R	NSL-KDD
Smart city	Rehman [47]	Ensemble ML	~97%	N/R	N/R	N/R	Enterprise IoT

Table 3. Cont.

Domain	Study	ML Approach	Acc.	FPR	Latency	Resource/Energy	Dataset
Healthcare	Naik [39]	CNN-LSTM-VAE + TAC	F_1 0.943	3.7%	<160 ms	RPi 5/Jetson Nano	H-IoT
Healthcare	Khan [26]	Autoencoder anomaly	>95%	N/R	N/R	N/R	General IoT
Healthcare	Karunamurthy [40]	Federated Learning IDS	~96%	N/R	N/R	Distributed edge	CIC/custom
Industrial	Aljabri [52]	BiGRU + Multi-Head Attention	98.28–99.74%	N/R	5.06 s/sample	N/R	CICIDS2018
Industrial	Kilichev [57]	CNN-LSTM-GRU IDS	>98%	N/R	14,800 s training	GPU only	EVCS
Industrial	Kantharaju [58]	SAPGAN/scalable ML	97–99%	N/R	Real-time capable	N/R	IIoT traffic
Industrial	Karthikeyan [55]	Firefly-optimized ensemble	99.34%	N/R	N/R	N/R	NSL-KDD
Industrial	Sagu [54]	GRU-CNN + SUCMO	96.65–99.1%	N/R	N/R	N/R	UNSW/BoT-IoT

3.5. Cross-Domain Comparison

Table 4 synthesizes the domain-specific analysis, revealing that optimal ML paradigms are context-dependent: lightweight classical approaches suffice for resource-constrained consumer IoT, federated architectures enable privacy-preserving deployment in distributed municipal systems, hybrid multi-objective frameworks address competing demands in healthcare and attention-based ensemble methods provide temporal modeling for critical infrastructure protection. Current domain-agnostic benchmarking, often reporting >99% accuracy on standardized datasets such as NSL-KDD, obscures critical deployment constraints. The field must transition toward domain-aware solutions that explicitly account for sector-specific requirements, validating against actual threat models, regulatory frameworks and operational realities.

Table 4. Domain-specific ML security solutions for IoT environments: threats, requirements, performance and deployment characteristics.

Focus Area	Primary Threats	Security Requirements	Optimal ML Paradigm	Accuracy Range	Reported FPR/Latency	Limitations	Key Studies
Smart home	DDoS (botnet recruitment), unauthorized access (brute-force), IoT-to-IoT attacks	- Privacy preservation - User interpretability - Edge-feasible (<1 s latency) - Multi-protocol support	Classical ML (RF, XGBoost) + XAI (SHAP/LIME)	97–100%	FPR: N/R Latency: 0.01–0.57 s Model: 2.73–1510 kB	Limited to known attacks; cross-device validation gaps	[30,33,35]
Smart city	DDoS (infrastructure disruption), large-scale attacks, unauthorized access	- GDPR-compliant FL - Scalability (millions of devices) - Real-time response	Federated CNN with robust aggregation (FedMADE)	94–98%	FPR: N/R Latency: 1.4 ms/sample (RPi 5)	FL maturity gap (22/104 use actual libraries); artificial partitioning	[18,31,38]
Healthcare IoT	Data tampering (vital signs), unauthorized device access, DDoS (monitoring disruption)	- Multi-objective security - Explainability (HIPAA/FDA) - Real-time (<160 ms) - FL for cross-institution	Hybrid CNN-LSTM-VAE + Trust-Aware Controller + XAI	94–99% (F_1 0.928–0.943)	FPR: 3.7% [39] Latency: edge-feasible (RPi 5, Jetson Nano)	Medical dataset scarcity; FL privacy proofs absent	[26,39,40]
Industrial IoT	APT (19% of breaches), ransomware (24-day downtime), SQL injection, XSS	- Adversarial robustness - Protocol-aware (Modbus/SCADA) - Real-time detection	Attention-based (BiGRU-MHA) or Ensemble (CNN-LSTM-GRU) with metaheuristic optimization	98–100%	FPR: N/R Latency: N/R (training: ~14,800 s for full ensemble [57])	OT network validation gap; no reported FPR in most studies	[52,57,58]

4. ML Techniques for IoT Security

This section synthesizes how different ML paradigms are applied to IoT security, focusing on empirical performance evidence from the reviewed literature rather than algorithmic fundamentals.

4.1. Classical ML Approaches

4.1.1. Supervised Learning

Supervised learning excels where labeled IoT attack datasets enable threat classification. Benchmark datasets such as CICIoT2023 [29] have facilitated models distinguishing DDoS, injection and unauthorized access with high accuracy on resource-constrained devices.

Among classical algorithms, Decision Trees achieve 99.56% accuracy on large-scale IoT datasets [49] with interpretability and low computational overhead suitable for edge deployment [34]. Random Forests demonstrate the most consistent performance across scenarios, exceeding 98–99% accuracy [26,49,59], with feature selection achieving 99.88% accuracy while maintaining 32-s training times [35]. SVMs optimized with firefly feature selection reach 99.34% accuracy [55], though scalability challenges in large-scale deployments often favor Random Forests for resource-constrained devices [51].

4.1.2. Unsupervised Learning

Unsupervised methods are essential for IoT environments where manual labeling is impractical and pre-labeled datasets lack representation of rare events [16,19,51,56,60]. These approaches detect zero-day attacks by identifying deviations from baseline behaviors without requiring labeled training data.

K-Means clustering serves as an efficient preprocessing step in IoT IDS pipelines, grouping normal traffic patterns to isolate anomalous clusters [30]. DBSCAN offers advantages by not requiring predetermination of cluster count, proving effective in federated learning environments for identifying outlier model updates from compromised nodes [38]. PCA enables dimensionality reduction for multi-modal IoT sensor data, integrated into deep learning pipelines to reduce feature space before classification [53]. Isolation Forest provides effective baseline anomaly detection in federated IoT IDS without labeled data, making it well suited to streaming environments [51].

Unsupervised methods typically exhibit higher false positive rates than supervised approaches, requiring integration with supervised classifiers in hybrid architectures [30,51]. In industrial contexts, they serve as first-line defense mechanisms flagging anomalies for subsequent review [16,60].

4.2. Reinforcement Learning

Reinforcement learning enables adaptive defense mechanisms where agents adjust security policies through environmental feedback [16,18,61]. RL agents dynamically adjust intrusion detection thresholds, balancing detection rates against false alarms [16,61], and autonomously modify firewall rules based on observed attack patterns, reducing response times from hours to milliseconds in industrial IoT environments [18]. These adaptive capabilities are valuable for zero-day scenarios where pre-programmed rules are ineffective.

RL also enhances operational efficiency by orchestrating bandwidth in wireless sensor networks [55,60] and scheduling edge computing resources to reduce latency [30,31,39]. However, current IoT-specific RL research remains limited in empirical validation, with most studies focusing on simulation-based evaluations rather than production deployments [61].

4.3. Deep Learning

Deep learning architectures excel at extracting hierarchical representations from high-dimensional IoT data, eliminating manual feature engineering requirements that are critical for heterogeneous environments [26,30,39,62].

4.3.1. Intrusion Detection

LSTM networks capture temporal dependencies in sequential IoT data streams. Hybrid CNN-LSTM architectures achieve F1 scores exceeding 0.94 with inference latency below 160 ms in healthcare IoT [39], while LSTM ensembles surpass 96% accuracy in multiclass IIoT attack detection [53,57]. VAEs provide unsupervised anomaly detection through reconstruction error analysis, achieving tamper detection $F_1 \approx 0.928$ (TPR 94.6%, FPR 3.7%) in healthcare deployments [39].

CNNs achieve test accuracies exceeding 99% on benchmark IoT datasets through spatial feature extraction from traffic representations [32,62,63], with federated CNN architectures reaching 95.59% accuracy while preserving data locality [40].

Hybrid frameworks combining CNN-GRU achieve 99.39% accuracy with 8ms latency, while BiGRU-MHA architectures attain 98.28% with 5-s execution times [33,52,54,57]. Ensemble approaches synthesizing CNN, LSTM and GRU achieve perfect binary classification and exceed 96% in multiclass IIoT scenarios [57].

4.3.2. Malware Detection and Authentication

SAPGAN architectures demonstrate 14–27% AUC improvements over conventional baselines across keylogging, exfiltration and DDoS variants [58], while ensemble methods with fuzzy logic provide graduated risk scoring [34].

Trust-aware adaptive controllers powered by CNN-LSTM-VAE achieve 96.1% accuracy in healthcare IoT access control through dynamic trust scores derived from anomaly detection and integrity assessments [39]. Unlike one-time authentication, behavioral biometric approaches enable continuous identity verification critical for medical and industrial IoT sessions [39].

Explainability techniques including SHAP and LIME provide transparency for CNN, DNN and TabNet architectures, facilitating informed security analyst decision-making [16,30,32,53,61].

4.4. Federated Learning

Federated learning enables collaborative model training across distributed IoT nodes through model parameter exchange rather than raw data transmission, preserving data locality [31,38,40,51,64].

Privacy is enhanced as raw data never leaves originating devices, mitigating breach and surveillance risks critical for healthcare, smart home and industrial domains [26,39,51]. Additional mechanisms including differential privacy, secure aggregation and homomorphic encryption provide formal privacy guarantees [31,40,51].

Federated IDS implementations have trained models across 10–150 clients with 94–98% detection accuracy [31,40,51], though heterogeneous data distributions require adaptive aggregation strategies [18,38]. Bandwidth efficiency is substantial: per-node memory drops from 2919 MB (centralized) to 542–819 MB (distributed across 5–10 nodes) [51], with aggregation adding approximately 4.7% latency overhead [38].

More than 100 studies employ FedAvg and related algorithms across CNN, DNN, LSTM and SVM architectures [18]. Dynamic aggregation methods achieve up to 71% accuracy gains for minority attack classes while maintaining poisoning resilience [38]. However, systematic analyses reveal challenges including reliance on legacy datasets with unrealistic partitions, limited robust aggregation against adversarial clients and insufficient energy profiling [18].

5. Machine Learning Techniques for IoT Security: Comparative Analysis

The proliferation of machine learning techniques in IoT security represents a transition from theoretical exploration to operational deployment, driven by the rapid expansion of connected devices, from 18.8 billion in 2024 to a projected 40 billion by 2031 [1], which generates attack surfaces of unprecedented scale that exceed the capacity of manual defense mechanisms. Notwithstanding the frequently reported detection accuracies exceeding 99% in controlled experimental settings [19,35,49,59], these performance metrics often obscure critical deployment considerations that fundamentally determine practical applicability: computational resource constraints, explainability requirements, privacy preservation mandates and adversarial robustness guarantees. This comparative analysis goes beyond the technique-only enumerations of [15,16] by binding every performance claim to the deployment constraint that justifies—or undermines—it.

This section departs from exhaustive technical enumeration to provide critical comparative analysis of machine learning approaches for IoT intrusion detection. We systematically present:

- a technique-focused taxonomy accompanied by quantitative performance profiles across representative deployment scenarios
- synthesized comparative insights that elucidate when deep learning architectures justify their computational complexity, the divergence between federated learning's theoretical promise and operational reality and the fundamental tension between model performance and interpretability, the explainability paradox
- an examination of persistent deployment challenges that impede the translation of laboratory results to production environments

Our comparative analysis demonstrates that classical machine learning approaches consistently match deep-learning accuracy on standardized benchmarks [19,26,35,49,59] at orders-of-magnitude lower training and inference cost (specific figures detailed in Section 5.2.1).

The advantages conferred by deep learning manifest in specific operational contexts: processing diverse data modalities in heterogeneous IoT deployments [32,62], modeling complex temporal dependencies in network traffic [33,54,57] and addressing multi-objective security requirements in critical applications such as healthcare IoT [39].

Analogously, while federated learning architectures provide essential privacy preservation mechanisms for distributed IoT deployments [31,38], a critical gap persists between theoretical architectural proposals and production-ready implementations (quantified in Section 5.3.5 and analyzed in detail below).

5.1. Technical Taxonomy and Performance Overview

Machine learning approaches for IoT security span classical algorithms, deep learning architectures, federated paradigms and hybrid frameworks. This subsection provides a high-level taxonomy with quantitative performance profiles; detailed comparative analysis follows in the next subsection.

5.1.1. Classical Machine Learning

Random Forest and Decision Trees consistently achieve 99%+ accuracy across IoT datasets [19,26,29,34,35,49,59] while maintaining computational efficiency suitable for edge gateways. Random Forest emerges as the most robust classical approach in the surveyed literature, achieving 99.88% accuracy with 32 s training time and 18-feature dimensionality reduction (down from 41 features) without accuracy loss [35]. Decision Trees with Grid-SearchCV hyperparameter tuning achieve 99.56% accuracy with minimal computational overhead and $O(\log n)$ inference complexity, enabling deployment on resource-constrained

microcontrollers [49]. Bio-inspired feature selection (firefly, grey wolf) lifts SVM accuracy to 99.34% in WSN-class IoT [55]. Training complexity and support-vector memory, however, limit scaling to high-dimensional traffic. In federated settings, Devine et al. [51] match Random Forest and ANN accuracy while sharply cutting per-node memory—though the most constrained edge devices remain out of reach.

5.1.2. Deep Learning–Convolutional Neural Networks

CNNs excel at spatial feature extraction from network traffic patterns, achieving up to 100% accuracy on device-specific datasets through automatic learning of hierarchical representations [32]. However, performance degrades substantially on heterogeneous, real-world traffic: 2.31% accuracy on the IoT-23 dataset encompassing Mirai, Okiru and Torii botnet variants across 23 diverse capture scenarios [62], versus 99–100% on controlled single-device datasets [32,33]. This heterogeneity sensitivity suggests CNN advantages manifest primarily in homogeneous deployment contexts rather than universally. CNN inference latency ranges from 0.01 to 0.57 s with model sizes 2.73–1510 KB [30,32], enabling edge deployment on consumer hardware (Jetson Nano, Raspberry Pi).

5.1.3. Deep Learning–Recurrent Neural Networks

LSTM and GRU architectures capture temporal dependencies in sequential network traffic, essential for detecting multi-stage attacks unfolding over time. Hybrid CNN-GRU approaches achieve 99.39% accuracy with 8 ms inference time, significantly faster than the CNN-LSTM baseline (28 ms) evaluated in the same study [33], while maintaining comparable accuracy through GRU's simplified gating mechanisms. Full ensemble architectures (CNN-LSTM-GRU) combining spatial feature extraction, long-term temporal memory and computational efficiency reach 100% binary accuracy and 96.90% for 15-class multi-attack scenarios [57], though training overhead (14,800 s for 50 epochs) necessitates GPU acceleration.

5.1.4. Deep Learning–Attention Mechanisms

Attention-based models (BiGRU with Multi-Head Attention) achieve 98.28% accuracy with sophisticated temporal pattern recognition through dynamic focus on salient traffic features [52]. Multi-head attention performs scaled dot-product attention across multiple representation subspaces, capturing diverse feature interactions (packet sizes, inter-arrival times, protocol distributions) that single-path architectures overlook. However, execution time analysis reveals attention mechanisms require 5.06 s per sample [52] versus 8 ms for CNN-GRU [33], representing a 632× latency increase for approximately 1% accuracy improvement, justified primarily for adversarial threat models (APT targeting industrial infrastructure) rather than commodity attacks.

5.1.5. Federated Learning

Federated learning enables privacy-preserving distributed training across geographically dispersed IoT devices through model parameter exchange rather than raw data aggregation. CNN architectures in federated settings achieve 94–98% accuracy across 10–150 clients with edge-feasible performance (1.4 ms/sample latency, <72 °C thermal operation on Raspberry Pi 5) [31]. Dynamic aggregation methods (FedMADE) address class imbalance through class probability matrix analysis and DBSCAN clustering, improving minority attack detection by up to 71% while introducing minimal communication overhead (4.7% latency increase, ≈55% s per training round) [38].

5.1.6. Hybrid and Ensemble Approaches

Multi-architecture integration leverages complementary strengths to surpass individual component performance. CTGAN-enhanced Random Forest pairs synthetic minority oversampling with native RF interpretability and post hoc SHAP/LIME explainability. The combination reaches 97–100% accuracy with edge-feasible deployment (0.01–0.57 s inference, 2.73–1510 KB models) [30]. Ensemble RF-BPNN couples RF robustness with BPNN non-linear decision boundaries and AdaBoost meta-learning. SDN/NFV orchestration enables automated response, yielding 99.9% accuracy [19]. Self-Attention Progressive GANs demonstrate 18–30% accuracy improvements over baseline CNN/DNN approaches through adversarial training for data augmentation and self-attention for long-range dependency capture [58], though generative training overhead limits practical applicability.

Table 5 consolidates the performance characteristics of the ML techniques surveyed in this section, enabling direct comparison across accuracy, computational cost and edge suitability dimensions.

Table 5. Summary of ML techniques for IoT security: performance and deployment characteristics.

Technique	Category	Accuracy Range	Training Cost	Inference Latency	Edge Suitable	Key Application
Random Forest	Classical	99.55–99.88%	Low (32 s)	Low	Yes	DDoS detection [35]
Decision Tree	Classical	99.56%	Low	Very low	Yes	Resource-constrained IDS [49]
SVM	Classical	99.34%	Medium	Medium	Partial	WSN intrusion detection [55]
Federated SVM	Classical/FL	Comparable to RF	Low	Low	Partial	Distributed IoT IDS [51]
1-D CNN	Deep	72–100%	Medium	0.01–0.57 s	Yes	Traffic classification [32]
CNN-GRU	Deep/Hybrid	99.39%	High	8 ms	Yes	Smart home IDS [33]
CNN-LSTM-GRU	Deep/Hybrid	96.90–100%	Very high (14,800 s)	Low	GPU only	Multi-class IDS [57]
BiGRU-MHA	Deep/Attention	98.28%	High	5.06 s	No	IIoT security [52]
GRU-CNN+SUCMO	Deep/Hybrid	96.65%	High	Medium	Partial	IoT anomaly detection [54]
Federated CNN	Deep/FL	94–98%	Medium	1.4 ms	Yes	Smart city IDS [31]
FedMADE	Deep/FL	+71% minority	Medium	+4.7% overhead	Yes	Imbalanced FL-IDS [38]
CTGAN+RF	Hybrid	97–100%	Medium	0.01–0.57 s	Yes	Augmented IDS [30]
SAPGAN	Hybrid/GAN	+18–30% over CNN	Very high	High	No	Data augmentation [58]

5.2. Critical Comparative Analysis

5.2.1. Deep Learning Complexity Under Theoretical and Practical Constraints

Performance analysis across over 25 studies reveals nuanced accuracy–complexity trade-offs that challenge the deep learning approaches. Classical ML (Random Forest, Decision Trees) consistently achieves over 99% accuracy on balanced, controlled IoT datasets [19,26,29,34,35,49,59], matching or exceeding deep learning performance while requiring orders of magnitude fewer computational resources: Random Forest trains in 32 s [35] versus CNN-LSTM-GRU ensemble requiring 14,800 s for 50 training epochs [57], representing a 462× training time difference. Similarly, inference latency comparison reveals Random Forest operates at the 32 s training scale with immediate prediction versus BiGRU-MHA attention mechanisms requiring 5.06 s per inference [52], constituting a 632× latency penalty compared to CNN-GRU’s 8 ms inference [33].

Deep learning advantages manifest primarily in three main scenarios:

- **Heterogeneous traffic patterns**
CNN architectures achieve up to 100% accuracy on device-specific datasets (individual smart home devices such as Philips baby monitors, GPS trackers and thermostats) [32] and 99.39% on single-device botnet telemetry [33], but degrade catastrophically to 72.31% on diverse, real-world IoT-23 traffic encompassing Mirai, Okiru and Torii botnet variants across 23 heterogeneous capture scenarios [62]. This performance

degradation suggests deep learning benefits are dataset-specific rather than universal, with advantages concentrated in homogeneous deployments where CNNs can learn device-specific traffic fingerprints through spatial feature extraction. Classical ML demonstrates more consistent performance across heterogeneity: Random Forest maintains 99.39–99.88% accuracy across diverse datasets (NSL-KDD, BoTNeT-IoT-L01, CICIoT2023) [26,35,49,59].

- **Temporal attack sequences**
Recurrent architectures (LSTM, GRU) capture multi-stage attacks unfolding over time reconnaissance, lateral movement, payload deployment—occurring in temporally separated phases that classical ML cannot model effectively. Hybrid CNN-GRU approaches achieve 99.39% accuracy on N-BaIoT [33], while GRU-CNN optimized via SUCMO achieves 98.71% on BoT-IoT [54], outperforming classical ML on attacks exhibiting strong temporal dependencies where feature-engineered approaches struggle to capture sequential patterns. However, for stationary threats without multi-stage temporal characteristics (brute-force authentication, isolated DDoS floods), classical ML matches or exceeds recurrent architecture performance without the computational overhead [35,49].
- **Multi-objective security**
Healthcare IoT requires simultaneous anomaly detection, integrity verification and access control, multiple security objectives that cannot be addressed through independent classical ML models owing to inconsistent alert thresholds and conflicting decision boundaries. Hybrid CNN-LSTM-VAE architectures achieve multi-objective security: anomaly detection ($F_1 \approx 0.943$), integrity verification ($F_1 \approx 0.928$) and access control accuracy of 96.1% through unified frameworks coordinated by Trust-Aware Controllers [39]. Classical ML cannot provide this integrated multi-objective capability, justifying deep learning complexity for healthcare contexts.

The literature suggests deep learning use is justified when heterogeneous traffic patterns require automatic feature extraction beyond manual engineering, temporal modeling captures multi-stage attack sequences or multi-objective security demands integrated frameworks. For stationary, well-characterized threats in homogeneous environments, classical approaches offer superior deployability (99.88% accuracy, 32 s training [35]) versus marginal deep learning improvements (98.28% accuracy, 5.06 s inference [52]).

Scenario-method fit across IoT domains.

The domain-by-domain rationale for these method-selection patterns is developed in the constraint-envelope analysis embedded in each subsection of Section 3.

5.2.2. Federated Learning Between Theory and Practice

A systematic review of 104 FL-IDS studies [18] reveals a critical implementation gap: only 22 employ actual FL libraries (FedML, Flower, PySyft), indicating that 79% of federated research consists of simulated partitioning rather than genuine distributed implementations. This maturity gap manifests in unrealistic client partitioning (random IID splits masking non-IID convergence challenges) [18,31], convergence overhead requiring hundreds of communication rounds under heterogeneous data distributions [31] and missing formal privacy mechanisms despite privacy being FL's core motivation [18].

Production-oriented implementations demonstrate FL viability: Federated CNN on Raspberry Pi 5 achieves 94–98% accuracy with 1.4 ms/sample latency [31], FedMADE achieves 71% minority attack detection improvement with 4.7% communication overhead [38] and Federated SVM reduces per-node memory from 2919 MB to 542–819 MB [51]. FL remains essential for privacy-critical domains but requires production-ready libraries with realistic partitioning, robust aggregation and formal privacy guarantees [18,38,51].

Domain-specific FL deployment maturity varies substantially. Smart cities have the strongest architectural need (GDPR-mandated distributed training, municipal data sovereignty) and the most active production research (Federated CNN on RPi 5 achieving 1.4 ms/sample [31]), but the 22/104 library-adoption rate reveals most city-scale FL proposals remain theoretical. Healthcare IoT has equally strong regulatory drivers (HIPAA prohibits cross-institutional data sharing) but faces an additional barrier: formal differential privacy proofs are largely absent from FL-IDS literature [18], meaning HIPAA compliance cannot yet be formally guaranteed. Industrial IoT presents the highest FL deployment barrier: OT network air-gaps and vendor certification requirements make even simulated multi-node federation unrealistic for most production environments; federated approaches in IIoT literature typically assume connectivity that operational networks deliberately prohibit. Smart home FL is technically feasible (residential gateways can participate as FL clients) but largely unnecessary: local inference on gateway-class hardware is sufficient for the threat models present in home environments, and privacy concerns are adequately addressed by on-device model execution without distributed training overhead.

5.2.3. The Paradox of Explainability in Machine Learning Systems

High-performing deep learning models achieve 98–100% accuracy [32,39,52,57] but operate as opaque black boxes providing predictions without explanatory rationale, conflicting with regulatory requirements (HIPAA medical device approval, FDA auditing, GDPR right to explanation), operational needs (forensic investigation, incident response, analyst trust-building) and stakeholder demands (clinical decision support, municipal transparency).

This explainability paradox, superior accuracy with inferior interpretability, creates deployment barriers in regulated, high-stakes domains. Post hoc explainability techniques address this partially through SHAP, which provides game-theoretic feature importance quantifying each input's contribution to predictions and LIME, which approximates complex model decisions locally through interpretable linear surrogate models [16,30,32]. However, these techniques introduce computational overhead (SHAP requires combinatorial feature evaluation scaling exponentially with dimensionality) and provide approximations rather than inherent interpretability; the explanations describe model behavior without guaranteeing faithfulness to actual decision logic, potentially misleading analysts when surrogate approximations diverge from true model reasoning.

Explainable-by-design approaches demonstrate that interpretability need not sacrifice performance. CTGAN-enhanced Random Forest achieves 97–100% accuracy with native interpretability plus SHAP/LIME transparency [30], while hybrid CNN-LSTM-VAE with Trust-Aware Controller achieves 96.1% access control accuracy with interpretable trust scores [39]. The literature suggests prioritizing inherently interpretable models or hybrid architectures with explainability modules rather than retrofitting XAI onto opaque models. When deep learning complexity is justified, explainability must be a first-class architectural concern, not a post hoc afterthought, as regulatory pathways increasingly require interpretable decision rationale [30,39].

The deployment case for XAI is strongest in healthcare IoT, where clinical audit requirements and FDA medical device approval pathways create mandatory interpretability obligations; SHAP-based feature attribution and Trust-Aware Controllers [39] directly address these regulatory drivers. In smart home environments, a practical interpretability need also exists: non-technical occupants making security decisions benefit from LIME-generated natural-language explanations, and the computational overhead of post hoc XAI on gateway hardware is acceptable. In contrast, smart city and industrial IoT deployments face a tension between XAI value and real-time constraints: SHAP's combinatorial feature evaluation does not fit sub-millisecond processing budgets in high-throughput OT or

municipal systems. For these domains, lightweight proxy metrics (feature importance rankings from Random Forest gini scores) offer a practical compromise, providing directional explainability without the full overhead of post hoc XAI methods.

5.3. Deployment Challenges

Despite remarkable laboratory performance (over 99% accuracy across numerous studies [19,26,32,35,49,52,57,59]), translating ML-based IoT security from controlled evaluations to operational deployments encounters persistent challenges spanning resource constraints, data quality, privacy preservation, scalability, evaluation methodology and operational maintenance. These barriers determine whether proposed solutions remain academic contributions or achieve practical impact.

5.3.1. Resource Constraints and Computational Efficiency

IoT devices exhibit severe computational limitations: microcontrollers with kilobytes of RAM, megahertz-class processors and milliwatt power budgets, orders of magnitude below requirements for executing sophisticated neural networks [35]. This resource scarcity creates fundamental tension between detection accuracy and deployment feasibility. Random Forest implementations with feature selection achieve 99.88% accuracy while maintaining edge feasibility (32 s training time, 18 features) [35], yet even these lightweight approaches assume gateway-class processors (Intel Core i5-1165G7 with 16 GB RAM) rather than endpoint sensors (ATmega328P microcontrollers with 2 KB RAM).

This computational gap forces hierarchical security architectures. Resource-intensive detection executes on edge gateways aggregating traffic from constrained endpoints [31,39]. The trade-offs are detection latency, single-point-of-failure risk if gateways are compromised, and bandwidth consumed transmitting traffic for centralized analysis. Edge optimization techniques (quantization, pruning, knowledge distillation) partially mitigate resource constraints, TensorRT quantization achieves 60% model size reduction with <1% accuracy degradation enabling CNN-LSTM-VAE deployment on Raspberry Pi 5 and Jetson Nano [39], but fundamental limits remain for battery-powered sensors requiring multi-year operational lifetimes.

The literature provides limited attention to energy-aware intrusion detection despite its criticality for battery-powered IoT deployments (environmental sensors, wearable medical devices, remote infrastructure monitors). Future research warrants development of energy-adaptive detection mechanisms dynamically adjusting model complexity based on remaining battery charge, prioritizing threat detection during critical operational phases while conserving energy during low-risk periods.

5.3.2. Data Quality Challenges in the Presence of Class Imbalance and Heterogeneity

Severe class imbalance where benign traffic comprises 90–99% of IoT network activity fundamentally challenges machine learning algorithms optimized for balanced datasets [26]. Extreme imbalance examples include CICIOT2023 where DDoS-ICMP flood contains 7.2 million samples while web-based uploading attack has only 1252 samples, a 5700:1 disparity causing standard classifiers to achieve high overall accuracy by correctly predicting the majority class while failing catastrophically on minority attacks [26]. SMOTE addresses this through k-nearest neighbor interpolation generating synthetic minority samples [26], CTGAN employs generative adversarial networks for realistic synthetic traffic generation [30] and weighted loss functions penalize minority class misclassifications more heavily during training.

However, synthetic oversampling introduces risks like overfitting to synthetic patterns not representative of real attacks, amplification of label noise where incorrect minority labels generate multiple synthetic errors, computational overhead limiting real-time applicability.

Validation on imbalanced test sets reflecting operational traffic distributions, not artificially balanced validation sets, remains essential for realistic performance assessment [26].

Data heterogeneity across IoT devices exacerbates imbalance challenges, as different device types generate vastly different traffic patterns rendering unified models suboptimal. FedMADE addresses this through Class Probability Matrix (CPM) dynamic aggregation clustering devices by traffic characteristics and adjusting aggregation weights based on local model contributions, achieving 71% improvement in minority attack classification [38]. However, this approach requires sufficient device diversity during training, potentially unavailable in specialized IoT deployments with homogeneous device populations (e.g., industrial sensor networks with identical hardware). Additional data quality issues include label noise from signature-based tagging or manual analysis creating training errors that propagate through models, temporal concept drift as IoT device firmware updates alter traffic patterns requiring continuous retraining [16,31] and adversarial contamination during federated training where attackers deliberately inject mislabeled samples to poison global models demanding robust aggregation mechanisms detecting and downweighting malicious clients [38].

5.3.3. Privacy Preservation Under Regulatory Compliance Requirements

Federated learning addresses privacy concerns by maintaining data locality; training occurs on distributed devices with only model parameters exchanged [16,18,31,38–40]. However, recent research reveals model parameter sharing can leak sensitive information through gradient inversion attacks reconstructing training data from shared updates without additional privacy protections [18].

Systematic FL-IDS review [18] identifies critical gaps:

- differential privacy providing formal information leakage bounds rarely implemented despite being essential,
- secure aggregation preventing servers from observing individual client updates largely absent,
- homomorphic encryption enabling computation on encrypted gradients prohibitively expensive for resource-constrained IoT.

Cross-border IoT deployments confront jurisdictional complexities where different regions impose conflicting privacy requirements. EU's GDPR prohibits data transfers to jurisdictions lacking adequate protection while mandating "right to explanation" for automated decisions. China's Cybersecurity Law mandates local data storage within national borders. US HIPAA imposes strict healthcare data protections with substantial penalties. These conflicting requirements potentially force region-specific detection models violating federated learning's assumption of global model sharing.

A persistent tension exists between privacy and threat intelligence: effective intrusion detection benefits from sharing attack indicators across organizations, but privacy regulations restrict data sharing without explicit consent. Differential privacy mechanisms adding calibrated noise to shared information enable privacy-preserving threat intelligence but degrade detection accuracy, the privacy–accuracy trade-off requires domain-specific calibration balancing regulatory compliance against operational effectiveness [18,40,51].

5.3.4. Scalability and Real-Time Processing

Real-time processing requirements mandate sub-second detection latency for time-critical applications (industrial process control, healthcare vital sign monitoring, autonomous vehicle networks) where delayed threat detection enables irreversible damage [31,32,39]. Healthcare IoT achieving sub-160 ms latency through edge optimization [39] assumes relatively low traffic volumes from individual medical devices; high-throughput IoT gateways aggregat-

ing traffic from thousands of sensors require processing gigabits per second, potentially exceeding even optimized model inference throughput.

Stream processing architectures address real-time requirements through incremental model updates and sliding window analysis, but introduce complexity (windowing strategies, late-arriving data handling, out-of-order processing) and potential detection gaps during model transitions [32]. The literature predominantly evaluates on static datasets using traditional train–test splits rather than streaming scenarios with continuous data arrival. This creates an evaluation–deployment gap: trained models may fail under operational streaming conditions exhibiting concept drift, time-of-day traffic variations and adversarial evasion attempts.

Hierarchical processing across edge, fog and cloud tiers promises scalability by mapping lightweight classical detection [35,49] to constrained endpoints, hybrid models [33] to gateways and attention-based deep analysis [52] to cloud. The cross-study view of how individual implementations score on scalability, energy efficiency and real-time feasibility together is given in the table from the Section 5.3.6 Optimal computation distribution, multi-tier failure handling and cross-tier threat-model consistency remain open challenges.

5.3.5. Dataset Limitations and Evaluation Methodology Gaps

Dataset quality fundamentally determines research impact, yet significant limitations plague existing IoT security datasets. Systematic review reveals NSL-KDD developed in 2009 remains the most frequently used dataset [18] despite being over 15 years old with outdated attack patterns (SQL injection variants from the 2000s), network characteristics diverging from contemporary IoT traffic (wired Ethernet vs. wireless protocols) and threat models predating modern IoT-specific attacks (botnet recruitment, IoT-to-IoT lateral movement). This persistence stems from community familiarity enabling cross-study comparison and established baselines, but perpetuates evaluation on unrealistic scenarios failing to reflect modern threats [18,29].

Table 6 compares the benchmark datasets referenced across the surveyed literature, highlighting their characteristics and known limitations.

Table 6. Benchmark datasets used in surveyed IoT security studies.

Dataset	Year	Attack Types	Environment	Used by	Known Limitations
NSL-KDD	2009	DoS, Probe, R2L, U2R	Wired enterprise network	[18,54]	Outdated attacks; no IoT protocols; wired only
UNSW-NB15	2015	9 attack families (Fuzzers, Exploits, etc.)	Synthetic + real network traffic	[54]	Limited IoT relevance; no device heterogeneity
BoT-IoT	2019	DDoS, DoS, recon, information theft	Simulated IoT network	[54]	Synthetic traffic; limited device types
IoT-23	2020	Mirai, Okiru, Torii botnet variants	Real IoT malware captures	[62]	23 capture scenarios; limited attack diversity
TON_IoT	2020	9 attack types	IoT/IIoT testbed with telemetry	[53]	Telemetry + network; moderate scale
AWID3	2021	Wi-Fi specific (deauth, evil twin)	802.11 wireless network	[36]	Single protocol (Wi-Fi); no multi-protocol IoT
Edge-IIoTset	2022	14 attack types (DDoS, injection, etc.)	IoT/IIoT edge devices	[52,57]	Laboratory environment; no operational OT traffic
CICIoT2023	2023	33 attack types across 7 categories	105 real IoT devices	[29,31,38]	Controlled lab; limited duration; no concept drift

Contemporary datasets (CICIoT2023 [29], Edge-IIoT [52,57]) address some limitations through realistic network topologies employing commercial IoT devices as both attackers and victims, but exhibit their own constraints:

- traffic captured in controlled laboratory environments rather than operational deployments, missing realistic background traffic, legitimate anomalies (firmware updates,

user behavior changes) and adversarial evasion techniques employed against known detection systems,

- limited duration (days or weeks) versus operational deployments spanning months or years experiencing concept drift,
- artificial attack generation using penetration testing tools rather than genuine adversarial campaigns exhibiting sophisticated anti-forensic techniques [29]

Most studies report accuracy, precision, recall and F1-score, but these metrics mislead when class imbalance or cost asymmetries characterize operational scenarios. False negatives (missed attacks) often carry far greater operational costs than false positives (false alarms), yet receive equal weighting in standard accuracy-based evaluation; cost-sensitive metrics should prioritize minimizing false negatives even at the expense of increased false positives [26,38]. Cross-dataset evaluation remains rare despite being essential for assessing generalization: X-FuseRLSTM evaluated across NSL-KDD, ToN_IoT and CICIoMT 2024 demonstrates cross-domain capability with 97–99% accuracy [53], but most studies train and test on single datasets using cross-validation, potentially overestimating real-world performance through dataset-specific overfitting.

The prevalence of accuracy $\geq 99\%$ across reviewed studies on NSL-KDD and CICIoT2023 warrants critical interpretation. These benchmark datasets exhibit structural properties that inflate reported performance: controlled laboratory capture conditions eliminate adversarial evasion, static class distributions prevent concept drift, and balanced evaluation splits mask the cost-asymmetry of false negatives versus false positives in operational security. Where FPR values are reported, as in the healthcare study by Naik et al. achieving 3.7% FPR [39], the operational implication is approximately one false alarm per 27 legitimate transactions, a rate that would overwhelm security analysts in real deployments. The field's near-universal omission of FPR, latency, and resource consumption metrics (documented across all four domain tables in Section 3) means that benchmark accuracy rankings do not reliably predict production deployment suitability.

The field would benefit from standardized evaluation protocols specifying mandatory cross-dataset validation (minimum 3 datasets from different sources), adversarial robustness testing (evasion attacks, poisoning attacks), resource profiling (latency, memory, energy measured on target hardware), concept drift analysis (model degradation over time) and cost-sensitive metrics (weighted F1-score, false negative rate as primary metrics). These protocols would significantly improve research reproducibility and practical applicability [18,29].

5.3.6. Deployment and Maintenance Challenges

Translating research prototypes to operational systems exposes gaps the literature only partially addresses. Concept drift as traffic patterns evolve requires continual model retraining, yet determining retraining frequency must balance model freshness (daily updates) against computational cost and potential instability from frequent changes [16,31]. Concept drift detectors triggering updates when performance degrades face reliability challenges: setting thresholds too conservatively causes unnecessary retraining overhead, while insufficiently sensitive thresholds allow degraded models to persist [31].

Federated learning promises decentralized model updates, but convergence can demand hundreds to thousands of communication rounds [31], stretching to weeks or months under intermittent IoT device connectivity. Asynchronous variants allowing clients to upload updates independently without synchronization barriers mitigate connectivity constraints but introduce staleness where some clients train on outdated global models, potentially hindering convergence [31,38]. Version control and rollback mechanisms prove essential when model updates introduce regressions (decreased accuracy, increased false

positives), yet federated settings complicate these safeguards as participants may operate on divergent model versions, maintaining global consistency while permitting local customization demands advanced distributed systems techniques seldom addressed in intrusion detection literature.

Operational deployments require security analysts to triage alerts, investigate incidents and provide feedback refining models. Active learning can reduce labeling costs and improve accuracy by querying analysts for the most informative examples (uncertain predictions, novel attack patterns), but mitigating analyst fatigue, designing effective query strategies and integrating these processes into security operations workflows demand interdisciplinary research spanning machine learning, human–computer interaction and security operations.

Cross-study deployment readiness.

Table 7 consolidates the per-study evidence on the three operational axes most often elided by accuracy-centric evaluation: scalability, energy efficiency and real-time feasibility. Ratings reflect the explicit evidence each source provides; “Not validated” indicates the dimension is not assessed rather than that it fails. The dominant pattern is asymmetric: most studies validate one axis at most, leaving practitioners without a multi-axis basis for method selection.

Table 7. Cross-study deployment readiness across scalability, energy efficiency and real-time feasibility. Ratings: H = high (validated evidence), M = medium (partial validation or simulated evidence), L = low (acknowledged limitation), NV = not validated. Each cell pairs a rating with a one-line evidence basis.

Study	Domain	Scalability	Energy Efficiency	Real-Time Feasibility	Evidence Basis
Albanbay [31]	Smart city	H (150 clients tested)	H (<72 °C, RPi 5)	H (1.4 ms/sample)	Hardware-validated FL
Sun [38]	Smart city	M (63 simulated clients)	NV	M (+4.7% overhead)	Sim-only partition
El-Sofany [19]	Smart city	NV	NV	NV	SDN/NFV concept
Naik [39]	Healthcare	NV (single-device study)	M (60% size cut)	H (<160 ms)	Edge-deployed RPi/Jetson
Khan [26]	Healthcare	NV	NV	NV	Lab-only autoencoder
Karunamurthy [40]	Healthcare	M (FL distributed)	NV	NV	Distributed-edge claim
Hasan [30]	Smart home	NV	M (compact 2.73 kB)	H (0.01–0.57 s)	Edge-feasible profile
Nawaz [35]	Smart home	NV	H (32 s training, 18 features)	H (low-cost RF inference)	Resource-profiled
Wang [33]	Smart home	NV	M (8 ms suggests low cost)	H (8 ms/sample)	Latency only
Aljabri [52]	Industrial	NV	L (5.06 s/sample)	L (5.06 s/sample)	Lab-only
Kilichev [57]	Industrial	NV	L (14,800 s training)	M (binary inference fast)	GPU only
Sagu [54]	Industrial	NV	NV	NV	Static benchmark

6. Domain-Specific Implementation Synthesis

This section synthesizes implementation evidence across four application domains, focusing on empirical findings, success factors, and research gaps. Table 8 quantifies performance–deployability trade-offs across domains. Whereas Ali et al. [17] catalogue domain risks without quantifying method–domain fit, the synthesis below reports accuracy alongside its primary deployability constraint for each domain, making the trade-off explicit.

Table 8. Quantified performance–deployability trade-offs of ML-based IoT security implementations across application domains (2023–2025).

Domain	Predominant Paradigm	Representative Accuracy Range	Primary Deployability Constraint	Core Performance–Deployability Trade-Off
Smart home	Classical ML + XAI (SHAP/LIME)	97.2–100%	Computational resources (model size 2.73–1510 kB)	Smallest models enable edge deployment but sacrifice F1 on minority classes; explainability adds minimal overhead
Smart city	Federated Learning (SVM, CNN, dynamic aggregation)	- F1 up to 0.981 - 98–99% accuracy	Communication overhead, data heterogeneity, non-IID distribution	Privacy preservation versus accuracy on minority attack classes; lightweight CNN offers best accuracy-efficiency trade-off on real hardware (RPi 5)
Healthcare IoT	Hybrid multi-objective DL (CNN+LSTM+VAE)	94.3–99.55%	Regulatory compliance, real-time requirements, multi-objective optimization	Feature reduction preserves accuracy while cutting latency; edge deployment (RPi, Jetson Nano) feasible but requires architectural complexity
Industrial IoT	Attention + ensemble + metaheuristic optimization	98.28–99.74%	Adversarial robustness, generalizability across datasets	Metaheuristic feature selection reduces input dimensionality (to 4–9 features) while maintaining >98% accuracy; attention mechanisms add robustness but increase model complexity

6.1. Smart Homes

Lightweight classical ML with post hoc explainability (SHAP, LIME) achieves 97–100% accuracy with edge-feasible characteristics: sub-second latency (0.01–0.57 s), compact models (2.73–1510 KB) suitable for consumer gateways [30]. Deep learning models (1-D CNN, DNN, TabNet) with XAI corroborate these findings, reporting up to 100% accuracy on IoT-specific sub-datasets [32]. CTGAN synthetic augmentation addresses class imbalance [30], while XGBoost feature selection reduces dimensionality from 41 to 18 features without accuracy loss [35].

Hybrid CNN-GRU offers marginal improvements (99.39%, 8 ms latency) [33], though classical ML achieving comparable performance (99.88%) with superior interpretability [35] suggests deep learning advantages remain limited for smart home contexts.

Key success factors include CTGAN augmentation maintaining statistical properties of attack traffic [30], feature importance-based dimensionality reduction [35] and SHAP/LIME integration bridging detection capability with user transparency [30,32]. Research gaps persist in multi-protocol frameworks (most studies evaluate single stacks) [36] and cross-device validation beyond specific benchmark configurations [29]. Future priorities include protocol-aware ensemble models, automated device profiling via transfer learning and privacy-preserving neighbourhood threat intelligence sharing.

6.2. Smart Cities

Federated learning architectures represent the predominant paradigm for privacy-preserving intrusion detection in distributed multi-stakeholder IoT environments [18,31,38].

Albanbay et al. [31] evaluate DNN, CNN and CNN+BiLSTM architectures within a federated environment simulating up to 150 IoT devices on CIIoT2023, achieving 94–98% accuracy with CNN and approximately 99% with CNN+BiLSTM. Edge deployment on Raspberry Pi 5 demonstrates 1.4 ms/sample inference latency and thermal op-

eration below 72 °C [31]. Sun et al. [38] introduce FedMADE, a dynamic aggregation method using Class Probability Matrix analysis with DBSCAN clustering. Evaluated with 63 simulated clients on CICIOT2023, FedMADE improves minority attack detection by 71.07% over FedAvg with only 4.7% latency overhead [38].

Key success factors include structured client partitioning reflecting operational heterogeneity [31], dynamic aggregation adjusting client contribution weights across class distributions [38], and hardware-validated edge profiling confirming commodity single-board computers can sustain real-time federated operation [31].

Limitations include artificial client partitioning using random data splits rather than operationally realistic distributions [18], missing privacy mechanisms (differential privacy, secure aggregation, homomorphic encryption) despite technical feasibility [18], and absent concept drift analysis validating long-term model stability [16]. Future priorities include production-ready FL frameworks with differential privacy integration ($\epsilon < 1.0, \delta < 10^{-5}$), standardized multi-stakeholder benchmarks, pilot deployments at operational scale and cross-organizational federated threat intelligence sharing [18].

6.3. Healthcare IoT

Hybrid deep learning configurations, particularly CNN-LSTM-VAE with Trust-Aware Adaptive Controllers, represent the most architecturally advanced approaches in healthcare IoT security [39], while lightweight classical ML with explainability achieves competitive accuracy [26,30].

Naik et al. [39] propose a trust-aware hybrid framework integrating CNN, LSTM and VAE components with a dynamic Trust-Aware Controller computing real-time trust scores from anomaly likelihood, context entropy and historical device behavior. The framework achieves 94.3% F1-score for anomaly detection and 96.1% access decision accuracy, with 12–18% improvement over rule-based baselines. Edge deployment through 8-bit quantization and TensorRT conversion achieves approximately 60% model size reduction with less than 1% accuracy degradation, enabling inference under 160 ms on Raspberry Pi 5 and Jetson Nano [39]. Khan and Alkhathami [26] achieve 99.55% accuracy using Random Forest with SMOTE oversampling, while Hasan and Tasnim [30] demonstrate 97–100% accuracy with CTGAN augmentation and SHAP/LIME explainability in edge-feasible model sizes (2.73–1510 KB).

Key success factors include trust-aware integration providing interpretable scores for clinical operators [39], edge optimization enabling commodity hardware deployment [39], and post hoc explainability essential for clinician-interpretable security decisions [30]. Limitations include medical-specific dataset scarcity, most studies employ general IoT benchmarks lacking healthcare traffic patterns (HL7 FHIR, DICOM) [26], and absent cross-institutional FL with formal privacy guarantees [18]. Future priorities include medical-specific threat models and datasets, standardized explainability interfaces for clinical decision support, privacy-preserving federated protocols, and EHR integration for context-aware security [39].

6.4. Industrial Internet of Things

Attention-based architectures and hybrid deep learning models represent recent approaches evaluated on IoT/IIoT-relevant datasets, though none have been validated in operational industrial environments [52,54,57].

BiGRU-MHA with SCSO-based feature selection and IWSO-driven hyperparameter tuning achieves 98.28% accuracy on Edge-IIoT [52]. CNN-LSTM-GRU hybrid architectures reach 100% binary accuracy and 96.90% for 15-class multi-attack scenarios on Edge-IIoTset for EV charging station security [57]. GRU-CNN models with SUCMO optimization

achieve approximately 96.65% accuracy on UNSW-NB15 and BoT-IoT [54]. Meta-heuristic optimization (IWSO, SCSO, SUCMO) automates model configuration, though the gap between binary (100%) and multi-class (96.90%) accuracy in [57] confirms that fine-grained attack discrimination remains challenging.

Key success factors include multi-head attention enabling diverse feature interactions [52], hybrid architectures leveraging complementary CNN/LSTM/GRU strengths [57] and meta-heuristic algorithms reducing manual tuning effort [54]. All reviewed studies evaluate on academic datasets rather than live OT networks with safety interlocks, real-time control demands and legacy protocol constraints [52,54,57]. No study validates against adversarial evasion techniques or industrial-specific attack scenarios [52] and the domain gap between academic datasets and operational IIoT environments targeting industrial protocols (Modbus, OPC-UA) remains unaddressed. Beyond threat detection, IIoT security increasingly encompasses operational integrity monitoring: AI-driven predictive maintenance [65,66] and IoT-based fault detection for industrial equipment [67] represent complementary security concerns in OT environments, where equipment failure and cyber-attack consequences are often indistinguishable. Future priorities include industry-specific threat models, operational OT datasets, adversarially robust models, physics-informed hybrid detection, and formal verification meeting IEC 62443 standards.

7. Conclusions

This review examined machine learning approaches to IoT security across four domains, smart homes, smart cities, healthcare and industrial IoT, synthesizing recent literature (2023–2025) to identify the current state of the field, persistent challenges and research trajectories toward operational deployment.

7.1. Summary of Findings

Classical ML approaches, particularly Random Forest and Decision Tree algorithms, consistently achieve accuracy rates exceeding 99% across multiple benchmarks, establishing their efficacy for resource-constrained environments [35,58]. Deep learning architectures capture complex attack patterns through hierarchical feature extraction, with hybrid models such as CNN-LSTM-GRU achieving 100% binary accuracy while maintaining low inference latency [57]. Federated learning addresses privacy preservation requirements but faces a substantial implementation gap: only 22 of 104 reviewed FL-IDS studies employ actual FL libraries [18], representing a critical barrier to real-world adoption.

Domain-specific analysis reveals distinct optimal approaches: smart homes benefit from lightweight, interpretable models with XAI integration (SHAP, LIME) [30,32]; smart cities require federated architectures with dynamic aggregation for multi-stakeholder scalability [31,38]; healthcare IoT demands multi-objective frameworks balancing detection, integrity and access control with edge-feasible inference [39]; and industrial IoT emphasizes attention-based architectures, though operational OT validation remains absent [52,57].

These findings directly validate the three contributions of this review. Contribution 1 (domain-specific mapping) is substantiated by the finding that no single ML paradigm dominates across domains: classical ML is optimal for smart homes, federated CNN for smart cities, hybrid DL for healthcare and attention-based ensemble for IIoT, with practical rationale grounded in threat profiles, resource constraints and regulatory requirements distinct to each domain. Contribution 2 (maturity-gap quantification) is evidenced by the FL library-adoption gap [18], the absence of FPR reporting across the reviewed studies (only 1 of 17 implementations in Table 3 reports FPR explicitly; see also Table 4), and the systematic documentation of accuracy inflation on benchmark datasets that do not reflect production conditions. Contribution 3 (practitioner roadmap) translates these findings into

time-horizon priorities: the immediate need for multi-metric evaluation frameworks and production-ready FL libraries, mid-term domain-specific dataset development and operational pilots, and long-term regulatory harmonization—addressing the specific deployment barriers identified in each domain rather than offering generic future work directions.

7.2. Positioning Relative to Existing Surveys and Robustness of Conclusions

Three existing surveys inform the positioning of this review. Alwahedi et al. [16] catalogue ML methods for IoT security without domain differentiation, providing technique-level breadth but no guidance on which paradigm is appropriate for a given deployment context. Ali et al. [17] present a broader IoT security taxonomy that encompasses both ML and non-ML defenses but does not quantify the gap between theoretical proposals and practical implementations. Alfahaid et al. [15] focus on ML security techniques for IoT with an emphasis on classification performance, again without distinguishing deployment-context requirements. This review advances those contributions by (1) mapping method selection to four concrete IoT domains with explicit rationale, (2) quantifying the documented FL implementation gap [18], and (3) providing time-horizon practitioner priorities rather than generic future work statements.

Regarding robustness, conclusions grounded in multiple studies across diverse datasets, such as the consistent superiority of classical ML for smart home contexts and the measurable FL communication overhead, are considered robust. Conclusions dependent on single studies or a single dataset warrant caution: the FPR of 3.7% for healthcare hybrid DL [39] derives from one study on one dataset; the CNN-LSTM-GRU training cost of 14,800 s [57] reflects a specific hardware configuration. Accuracy figures above 99% reported on NSL-KDD or CICIOT2023 should be interpreted as upper-bound estimates under controlled conditions rather than predictors of production performance, for reasons detailed in Section 5.3.5.

7.3. Open Challenges

Four fundamental challenges constrain practical deployment. First, IoT resource constraints (limited processing power, memory, energy) necessitate continuous trade-offs between detection accuracy and computational feasibility, with the optimal edge/cloud balance remaining context-dependent. Second, extreme class imbalance where malicious traffic comprises less than 1% of total activity challenges conventional algorithms; synthetic oversampling (SMOTE, CTGAN) shows laboratory efficacy but lacks real-world validation under concept drift [30]. Third, deep learning opacity poses operational challenges for security analysts; while post hoc XAI provides partial interpretability, inherently interpretable security-specific models remain underexplored [32]. Fourth, privacy preservation conflicts with centralized learning; federated approaches must address heterogeneous data distributions, communication overhead and poisoning vulnerabilities [18].

7.4. Research Roadmap

We propose a prioritized research roadmap organized into three implementation horizons.

Immediate priorities include: (1) a standardized cross-dataset evaluation framework mandating validation on minimum three diverse datasets with reporting of adversarial robustness, inference latency, energy consumption and memory footprint alongside accuracy; (2) production-ready FL frameworks employing actual libraries (FedML, Flower) with geographic partitioning, robust aggregation and differential privacy integration ($\epsilon < 1.0, \delta < 10^{-5}$); and (3) contemporary domain-specific datasets replacing over-reliance on NSL-KDD (2009) with realistic IoT traffic capturing device heterogeneity and modern attack vectors [18].

Mid-term priorities encompass: smart home multi-protocol ensemble detection with privacy-preserving neighborhood threat intelligence; healthcare regulatory alignment through medical-specific threat models, standardized explainability interfaces for clinical workflows and HIPAA-compliant federated protocols; industrial OT network validation with adversarially robust architectures and physics-based anomaly detection meeting IEC 62443 requirements; and smart city pilot deployments validating federated architectures at operational scale.

Long-term priorities address: cross-domain transfer learning reducing training data requirements in data-scarce domains; international regulatory harmonization through ISO/IEC JTC 1/SC 27 and NIST for ML-based IoT security certification with cross-border federated threat intelligence; and energy-aware hardware-optimized detection dynamically adjusting model complexity based on available energy budget.

7.5. Concluding Statement

As IoT deployments expand toward 40 billion connected devices by 2031, the urgency of developing robust, scalable security solutions intensifies. The evidence demonstrates that, while ML has achieved remarkable laboratory success, translating these advances into operational security requires addressing resource optimization, privacy preservation and system interpretability through coordinated effort among ML researchers, security practitioners, domain experts and regulatory bodies. This review advances existing surveys through three equally weighted contributions: systematic domain-specific requirement mapping demonstrating context-dependent method selection, quantification of the maturity gap between FL theory and practice, and a prioritized practitioner-oriented roadmap with time-horizoned implementation priorities. Together, these provide the analytical framework and actionable recommendations necessary to bridge the gap between promising benchmarks and trustworthy deployed solutions.

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Abbreviations

The following abbreviations are used in this manuscript:

ACK	Acknowledgment (TCP flag)
AWID3	Aegean Wi-Fi Intrusion Dataset
BDT	Boosted Decision Tree
BiGRU-MHA	Bidirectional GRU with Multi-Head Attention
CCTV	Closed-Circuit Television
CVE	Common Vulnerabilities and Exposures

CNN-GRU	Convolutional Neural Network-Gated Recurrent Unit
CTGAN	Conditional Tabular Generative Adversarial Network
DBN	Deep Belief Network
DDoS	Distributed Denial of Service
DL	Deep Learning
DNN	Deep Neural Network
DoS	Denial of Service
DQN	Deep Q-Network
DRL	Deep Reinforcement Learning
ECG	Electrocardiogram
FL	Federated learning
GRU	Gated Recurrent Unit
HIPAA	Health Insurance Portability and Accountability Act
HVAC	Heating, Ventilation and Air Conditioning
IDS	Intrusion detection system
IIoT	Industrial Internet of Things
IP	Internet Protocol
IoT	Internet of Things
IWSO	Improved White Shark Optimization
LIME	Local Interpretable Model-agnostic Explanations
LSTM	Long Short-Term Memory
MITM	Man-in-the-Middle
ML	Machine learning
MQTT	Message Queuing Telemetry Transport
NFV	Network Functions Virtualization
OSNN	Optimized Single Neural Network
RL	Reinforcement Learning
SAPGAN	Self-Attention Progressive GAN
SCSO	Sand Cat Swarm Optimization
SDN	Software-Defined Networking
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
SOHO	Small Office/Home Office
SUCMO	Self-Upgraded Cat and Mouse Optimization
SVM	Support Vector Machine
SYN	Synchronize (TCP flag)
TAC	Trust-Aware Adaptive Controller
VAE	Variational Autoencoder
XAI	Explainable Artificial Intelligence

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